Cryptocurrency, Stablecoins and Blockchain: Exploring digital money solutions for remittances and inclusive economies

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Abstract

For sizable populations in low- and middle-income countries (LMICs), remittances from abroad form significant portion of their income. The remittances are monies sent to them by their close relatives or friends who work as migrant workers in high income countries. The total remittance amounts have been steadily growing over the last several decades to hundreds of billions of dollars as of 2019 (World Bank). For some low-income countries, these funds account for as much as 10 to 15% of their annual GDP. As important as this source of income is for those living in LMICs, the transfer of money may involve untrusted third parties that charge hefty fees to facilitate such transfers. These fees can be anywhere from 5% to 10% of the total funds per transaction. Even if majority of these transactions occur without any problems, there is always possibility of fraud and theft resulting in loss of income for those who depend on such funds. The use of cryptocurrencies and blockchain technologies is an alternative to money transfers. Use of these technologies eliminate need for untrusted third parties in funds transfer operations. With increasing penetration of mobile phone technologies all, even those who may

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be “unbanked” are likely to have access to mobile phones. In such case, cryptocurrency/blockchain technology can be utilized for funds transfer to their mobile devices.

There are at least two well-known cryptocurrency companies (Ripple and Stellar) that help in near instantaneous digital funds transfer for a tiny fraction of fees charged by banking industry and other money transfer operators; although their global reach is limited to high-income countries. What is needed in the LMIC case is an entity that has global appeal and reach.

The recent announcement of a new cryptocurrency called “Libra” by Facebook may be one such candidate; given that Facebook has a vast social network spanning the entire globe with nearly 2.5 billion active users. That could help its cryptocurrency Libra as medium of exchange and cross-border funds transfer.

In this paper we consider remittance operations with and without cryptocurrency and blockchain technologies in general and present pros and cons of Libra for cross-border payments and money transfers. We also present a brief discussion on challenges posed by global cryptocurrencies to central bank governing bodies all over the world and policy implications arising out of potential conflicts between sovereign currencies and future of global cryptocurrencies such as Libra.

Key Words: Cryptocurrency, Blockchain, Stablecoin, Ripple, Stellar, Libra, Remittance, Inclusive Economy

Background

Over the last several decades, since the end of World War II, the developed world has provided aid in many forms, both public and private; bi-lateral and multi-lateral; consisting of finances, goods and services; targeted sectoral grants as well as loans offered at concessional rates.

According to OECD, between 2005 and 2015 nearly $2.9 trillion in aid flowed from high income countries to many LMICs, averaging over $260 billion per year (Chart 1).

Welcome as it is for many of its success stories, it is not without controversies both in the donor and recipient countries. (https://www.jstor.org/stable/24385619). There is a vast and growing literature on the use and abuse of "foreign aid." In this paper, we will not delve into pros and cons of international assistance provided by developed countries to the developing world. Instead we will discuss another form of aid, the "remittances", that have become very critical to many LMICs over the last several decades. Further, we will examine whether the highly innovative and dynamic new payment systems based on cryptocurrencies could help or hinder "remittance" processes and also can these new payment systems be a solution to "unbanked" population all over the globe (Map 1).

Remittances

Remittances are the no-strings-attached money transfers from the developed world to the developing world. These are vast number of small funds sent by expatriates to friends and families back home. Over the years, the size of remittances has grown steadily to over half a trillion dollars per year (Chart 2). At the aggregated levels, the total remittance amounts are larger than the total development aid flowing from developed world to developing world. Map 2 shows a stylized "from-to" remittance flow diagram.

For sizable populations living in LMI countries, remittances from abroad form a significant portion of their income. When aggregated at the country level, the remittances account for between 2 to 5 percent of LMIC's GDP (Map 3). And for a tiny minority of LMICs, these remittances account for 10% or more of their GDP (Table 1).

Network topology diagram in Fig-Net 1 shows interconnections between "donor" country nodes and "recipient" country nodes, where an (directed) edge exists if a "donor" country sends remittances to a "recipient" country and size of the node represents total remittance amounts. Fig-Net2 and Fig-Net3 show "donor" and "recipient" node view of the donor-recipient network and the size of a node represents the "remittance" amounts. Clearly the U.S.

and U.K. are the top donor countries while China and India are the top recipient countries. It is important to note that a top recipient country at the global level may be a top donor country locally. For example, India ranks nearly at the top as a remittance recipient country, but at the local level, it's a top donor country for Nepal, Bangladesh, Sri Lanka and Bhutan. Such regional corridors make some of these dominant recipient countries, top donor countries locally. Further, the transmittance fees vary from country to country and from one region to another.

As important as this source of income is for those living in LMICs, the transfer of money involves third parties. There are over 500 legitimate third parties such as banks, money transfer operators, post offices etc. who facilitate transfer of funds; the majority of which fall into one of the two main categories, banks (ANZ Bank, CitiBank, WestPac, WalMar2World etc.) or money transfer operators (Western Union, Money Gram, WorldRemit etc.). Money Transfer Operators (MTO) dominate the field with global spanning networks scattered across thousands of locations; and all they need is a legit ID from both the sender and the recipient to transfer funds. However, no matter which category they belong to, nearly all of them charge hefty fees to facilitate these cross-border transfers which range between 1% to as much as 20% with a global average of 7% (Map 2). Money transfer costs are higher in Sub-Saharan countries while countries in Middle-East and North Africa typically pay lower fees. Then there are countless number of unofficial money transfer channels such as Hawala, Fen Chui, Hundi, Chits etc. (IMG Guide: International Transactions In Remittances) that may offer reasonably reliable services at lower costs. While majority of these transactions occur without any problems, there can be significant delay in funds reaching their final destinations, as well as potential for fraud and theft resulting in loss of income for those who depend on these funds. In the present work, we consider if there a way to reduce fees, cut time delays and guarantee transfer between source and destination.

The answer may lie with the technologies of cryptocurrencies and blockchain. These could help to facilitate fast and guaranteed transfer of funds at fraction of cost currently charged by money transfer operators. In particular, the stablecoin cryptocurrencies could be the solution that will ensure cost-effective, guaranteed and timely transfer of funds from donors to recipients. The following section has a brief discussion on basics of blockchain and cryptocurrencies, followed by introduction to stablecoins, a type of cryptocurrency that is supported by a set of real-world liquid assets or a combination of liquid assets and a basket of fiat currencies. The Stablecoin section also has discussion on what constitutes a digital currency and examples of digital currencies and stablecoins including FaceBook's entry into the cryptocurrency world with a stablecoin called Libra, the cryptocurrency that wants to be world's global stablecoin. It also includes discussion on what political and regulatory hurdles Libra is facing and whether Libra controversy will spur development of digital currency equivalent of fiat currencies. According to some reports, Libra announcement may have influenced China's decision to pursue digital version of renminbi. Unlike private digital currencies, fiat currencies issued by sovereign governments are legal tender and could in theory take any form: cash, bank deposits or digital currency.

It is important to note that, many of the big and small tech and social network companies that are trying to create cryptocurrencies with their implementation of underlying blockchain infrastructure; they are – either knowingly or unknowingly – trying to be part of a well-established and largely functional banking and finance sector. These upstarts who are used to unregulated world of tech and social media sector that try to market their products anywhere in the world are suddenly entering a sector that is governed by decades old regulatory infrastructure. At the very least, they should be aware that they must comply with the KYC (Know Your Customer or Customer Due Diligence https://www.govinfo.gov/content/pkg/FR-2016-05-11/pdf/2016-10567.pdf) and AML (Anti-Money Laundering https://www.fincen.gov/history-anti-money-laundering-laws) regulations. In some cases these fintech wannabes with their newly minted digital currency may be challenging sovereign currencies across the world and thus may face at the least extra scrutiny on how they are governed and in many cases they may face outright ban (https://www.loc.gov/law/help/cryptocurrency/world-survey.php).

Cryptocurrencies and Blockchain

Typically, cryptocurrency (also referred to as digital currency or virtual currency) refers to digital entries in a distributed ledger (database) maintained by a network of nodes (computers). In theory, all peer nodes in a network have every record of all transactions since the time the said digital currency is born.

Each peer node is assigned a digital wallet that consists of two cryptographically generated keys, one a secret and private key represented here by symbol PrX and a public key represented by symbol PuX. Private key PrX is used to digitally sign each transaction initiated by that node/wallet, while the public key – shared with the world – is used to derive public address associated with that peer node.

Consider two peer nodes A and B, each has a private key (PrA and PrB) and a public key known to the world (PuA and PuB) along with their wallet address WA and WB derived from their respective public keys. Suppose node A wants to send k amount of digital currency to node B, then node (A) initiates a transfer by digitally signing the transaction with the secret private key PrA and broadcasts the transaction to the network with destination address WB derived from B's public key PuB. The actual transfer of digital funds does not take place unless the transaction is verified by the peer nodes. The verification involves checking that Node A has not initiated duplicate transactions and that node A has sufficient balance to cover the combined total of transaction amount plus a small fee[2] to process the transaction along with any other amounts and associated fees for pending (unverified) transactions from node A.

The distributed ledger is maintained as a cryptographically generated chain of blocks. As peer nodes transact with each other using cryptocurrency, these transactions are periodically compiled in to blocks and verified by any of the peer nodes and then appended to existing chain of blocks. Incentive for any of peer nodes to verify new transactions in timely manner is the award of "freshly mined" digital currency where mining refers to process of solving a cryptographic puzzle. Competing set of nodes try to solve the cryptographic puzzle by compiling variable number of new and unverified transactions into blocks. A predetermined number of new digital currency is awarded to a peer node that is the first one to solve a cryptographic puzzle after rest of the community of nodes reaches a consensus after verifying that the

[2] Cryptocurrency transaction fees are tiny and are independent of the transaction amounts. In other words, generally, fees stay same whether transacted amounts are small or very large.

cryptographic puzzle has indeed been solved. The cryptographic puzzle here refers to finding a hash value for a block (set) of new and unverified transactions compiled by a node (also referred to as miner) such that the hash value is smaller than a preset hash – also called difficulty level - set by the network. All the nodes in this network arrive at a consensus by verifying the solution and update their copy of the distributed ledger by appending (chaining) the "freshly minted" block to the existing chain of blocks aka Blockchain. The entire process of growing blockchain by appending cryptographically verified transaction blocks and reaching consensus on the state of the blockchain makes it highly secure and tamperproof[3].

The process described above is similar to how Bitcoin (https://bitcoin.org/bitcoin.pdf) and Ethereum (https://github.com/ethereum/wiki/wiki/White-Paper), the two of the world's largest blockchain based cryptocurrencies work. Since Bitcoin rose to be the world's first highly successful digital currency, hundreds of private sector cryptocurrencies have tried to duplicate Bitcoin kind of success. Not quite like bitcoin but to some degree, a few have been quite successful while others are barely surviving.

Nearly all these digital currencies suffer from very high degree of volatility (https://cci30.com). Sometimes in just in matter of hours value of a crypto currency could dip or rise substantially, making these poor candidates as store of value or medium of exchange. But compared to fiat currencies, associated transaction completion times and transaction fees are either on par or negligible. Once a transfer is complete, the receiving party can start using the digital currency immediately. However, the volatile nature of cryptocurrencies makes these digital currencies a poor choice for remittance operation. So are there any cryptocurrencies that are not as volatile, have small transaction fees and very little delay in transaction completion? And the

[3] If any of the peer nodes tries to change past transactions then that node will have to solve the cryptographic puzzle for that block as well as all of the subsequent blocks, a computationally very expensive and impossible to carry out such task for a single node or even for a group of nodes. Unless a majority of nodes combine their computational efforts and fork an existing blockchain creating blocks with transactions that are beneficial to them.

answer is a qualitative yes, and these types of cryptocurrencies are generically referred to as stablecoins.

Stablecoin

Stablecoin refers to a class of cryptocurrencies that are backed by a set of real-world liquid assets so that price stability of real-world assets is reflected into degree of stability associated with these cryptocurrencies[4]. Real world liquid assets could be a basket of currencies or a reserve[5] currency (deposited with central banks). The prime feature of Stablecoin backed by a reserve currency is that anyone who owns a stablecoin can convert it for a fiat currency at its face value at any of the online cryptocurrency exchanges where that stable coin is listed. For trading. For example, if party A has purchased a stablecoin **s** for **d** dollars then that party will be able to redeem the dollar value any time in the future. As the name indicates, these class of cryptocurrencies provide certainty of price stability that is lacking in other cryptocurrencies, a very desirable attribute for remittance: as a store of value and exchange of medium and unit of account.

Further, Stablecoin as a cryptocurrency inherits many desirable properties associated with cryptocurrencies: pseudo-anonymity, speed of transactions, very low transaction fees, security and safety. Additional desirable traits such as ease of use with mobile phones using social media networks, and global reach would make stablecoins ideal for remittances.

In brief, an ideal stablecoin has the following properties:

- Store of value guaranteeing long term stability,
- Medium of exchange that is used as cash for purchase of goods and services,
- Unit of account and? all economic activities can be measured and compared using a standard unit.

[4] Real world stable currency (US Dollar) or precious metal (Gold) pegged stablecoin is just one type. There are two other types of Stablecoins, one backed by overcollataralized another cryptocurrency and third one is non-collartaralized where the desired price stability is achieved through smart contracts. (Hackernoon article)
[5] US Dollar is World's reserve currency (https://en.wikipedia.org/wiki/Reserve_currency)

- It has global reach, i.e., can be transferred between any two or more parties with no geographic border constraints;
- The transfers are almost instantaneous (quick settlement of transactions),
- Such transfer cost next to nothing (very small transaction cost),
- The transfers are agonistic about the use of devices (Not tied to brand names),
- It is very easy to use, similar to texting on mobile devices,
- and with the added benefit of pseudo-anonymity at user's discretion (similar to cash).

So, are there any real world stablecoins? Not quite but there are some public and private digital currencies and cryptocurrencies that may fulfill, to various degrees, some of the desirable traits associated with stablecoins. There are several digital currencies, each of which operate within a country or in a case or two regionally where they are supported by that nation's phone (telecom) networks.

Alipay, WeChat and Renminbi

China's online commerce giant, Alibaba's Alipay, and China's social network company, TenCent's WeChat, are examples of online and mobile phone based digital payment services that are very widely[6] used in mainland China. Both have also been trying to set up cross-border payment services with neighboring countries. It is important to note that both Alipay and WeChat are highly regulated by China's Central Bank (equivalent of Federal Reserve in the U.S.) In fact, China's Central Bank has declared its plans for digital cash version of Renminbi and will become part of the M1 money[7]. Obviously both Alipay and WeChat have the potential to be cross-border payment services and to some extent they do that regionally, but they are tied to Renminbi. Unlike US Dollar, Renminbi is yet to be accepted as a reserve currency of the world.

M-Pesa

[6] With widespread internet penetration across China and adoption app-based electronic payment services, China is rapidly moving towards a cashless society. (https://dailyfintech.com/2019/11/01/china-the-age-of-bat-is-over-as-pinduoduo-taps-into-rural-ecommerce/)
[7] M1 money consists of coins/cash in circulation and bank deposits. https://www.federalreserve.gov/releases/h6/current/default.htm#t1tg1f1

Even before Bitcoin's birth, there was M-Pesa, a mobile phone based money transfer services that operated in Kenya. Even though it has grown into a regional digital currency of sorts for money transfer and digital payment services, it is far from becoming a stablecoin due to its limited use and reach.

Venezuela's Petro is a government sponsored digital currency backed by that country's petroleum reserves, however, due to its current status as a pariah state in geo-politics, it is unlikely to have any either as a cryptocurrency or a stablecoin beyond its borders.

Private Stablecoins

There exist couple notable examples of private enterprises working towards the goal of establishing their cryptocurrencies as alternative stablecoins that could be useful for cross-border payments.

Tether

Tether is a cryptocurrency that is pegged to U.S. dollar, however, Bitfinex (http://www.bitfinex.com/), the Hong Kong based cryptocurrency exchange that issues tether, has been dogged by various controversies, including doubts about size of its reserve currency (i.e., the U.S. dollar) to keep the 1:1 ratio between a tether coin and the U.S. dollar. A Recently published online paper (Griffin and Shams, 2019) claims that Bitfinex may have a role in using its stablecoin tether to drive up the prices for bitcoin to historic heights in late December, 2017 ($20,000 per bitcoin: https://fortune.com/2017/12/17/bitcoin-record-high-short-of-20000/). Obviously, tether/bitfinex appears to be less than ideal as a stabelcoin solution that could useful for remittances/cross-border payments.

XRP and XLM

For last several years Ripple's XRP (https://www.ripple.com/) and Stellar's XLM (https://www.stellar.org/) have been in use for funds transfer. Reportedly, a few international banking giants have been using Ripple's near real-time cross-border payment transfer technology called xRapid (https://decrypt.co/5313/complete-ripple-partnerships-xrapid-xrp) instead of SWIFT (https://www.swift.com/), the standard industry bearer for the banking sector. Ripple's digital network is able to settle funds transfer transactions in near real time that cost as low as just a few pennies or less. However, much of those inter-bank transfers have been done with digital equivalent of fiat currencies and not using Ripple's native digital currency, the XRP. Ripple's XRP has potential to be a stablecoin, however, over the last few years it has experienced moderate to high levels of volatility just like Bitcoin (https://bitcoin.org/en/) and Ether (https://ethereum.org/), the two largest cryptocurrencies in terms of market caps and many other cryptocurrencies.

USDC

USDC (https://www.coinbase.com/usdc) from the U.S. based cryptocurrency exchange Coinbase nearly fits the very definition of stablecoin as it is pegged to US Dollar: One USDC coin can be bought or sold for exactly $1.00 and with zero fees as long as one uses Coinbase accounts. USDC, an Ethereum based token, has been promoted as a programmable cryptocurrency, i.e., it can be part of a smart contract transaction code where price stability is required. Coinbase operates in over 80 countries and has the potential to be used as a cryptocurrency for a near real-time cross-border payments. However, similar to Ripple, both the sending and receiving parties have to have Coinbase accounts for cross-border transactions and the only conversion that is possible at either end is from USDC to US Dollar and back. Thus, those who may want to use USDC for cross-border funds transfer will still have to deal with local infrastructure involved in converting US Dollar to local fiat currencies, which is not an easy task, especially if you live in one of the low to middle-income countries.

Libra

Facebook (FB) announced its entry into cryptocurrency /blockchain field with the unveiling of Libra, the "stable cryptocurrency" and a "distributed" blockchain that is governed by not for profit "Libra Association" (Figure 6). According to Libra white paper (https://libra.org/en-US/wp-content/uploads/sites/23/2019/06/LibraWhitePaper_en_US.pdf), Libra will be a stablecoin backed by real world reserves in the form of a basket of fiat currencies and government securities. The founding members of Libra Association consist of "diverse group of organizations from around the world" (https://libra.org/en-US/association/#goals_organization). The members of Libra Association are involved in verifying transactions and blockchain, which are also referred to as "validator nodes."

In addition to Libra, Facebook has also created a subsidiary called Calibra that will develop financial and payment services rendering its version of cryptocurrency/digital wallets[8] called Calibra wallet. According to FB's Calibra website (https://calibra.com) it will be "a standalone app in the App Store and Google Play. You will also be able to use Calibra directly in your WhatsApp and Messenger apps, so you can send and receive money as easily as you message friends, family, and businesses." FB says, Calibra digital wallets (Fig 7) will be a "A New Digital Wallet for a New Digital Currency". (https://about.fb.com/news/2019/06/coming-in-2020-calibra/) The goal here is to use social network apps to make financial services as easy to use as sending a message using Messenger or WhatsApp and with very small transaction fees. FB thinks that with the increasing penetration of mobile phones where internet access is limited (Map 7); and there are no banking services, those with mobile phones could use Calibra digital wallets to send or receive money (Libra) and for buying goods and services. As per FB's estimates the combination of Libra/Calibra with its social network apps will be able to provide access to financial services to very large cohort of nearly 1.7 billion unbanked population across the globe.

Assume that both A and B have FB accounts which in theory may satisfy the KYC regulation.
Assuming there is a 1:1 parity between dollar and Libra, person A buys 100 Libra at a cryptocurrency exchange for $100 plus a small fee charged by the exchange.
Libra Association will create 100 Libra backed by $100.
Person A then uses her Calibra app to send 100 Libra to her family B's Calibra app on their FB account for a small transfer fee.
Family B is then able to either use Libra to buy goods and services locally or convert it to local currency X or convert it into back into $100.

[8] Typically, digital wallets hold secret digital keys that are used to digitally sign transactions for sending or receiving cryptocurrencies. These may also hold the addresses on the blockchain ledger of digital assets. Wallets with keys and ledger addresses help one prove ownership of digital assets.

In the latter? Case, equivalent amount of Libra will be burnt (electronically destroyed)
Each of these transactions are verified and added to Libra blockchain.

Combination of FB's global reach of its social network platforms with embedded financial services applications from Calibra, and its stablecoin Libra, would be an ideal solution for cross-border payments/ remittances. But Libra won't be operational till the latter half of 2020. However, technology is not the biggest hurdle for Libra. Libra faces political and regulatory obstacles, which may not be overcome.

FB does not have to deal with any compliance issues when it comes to its use of social media applications all over the world. However, when those applications involve financial transactions, Libra/Calibra may have to have separate agreements with finance and banking regulators in each of the countries where it wants Libra to be recognized as a stablecoin. Already, regulators in many countries have sounded alarm over Libra as a potential threat to their sovereign currencies. Finance ministers in Eurozone countries (France, Germany, Italy, Netherlands) have said that they would actively oppose Libra (https://www.politico.eu/article/facebook-libra-faces-eurozone-backlash/). Further, the U.S. treasury department has raised issues about how Libra might run foul with AML (Anti Money Laundering) regulations and could be used by terrorist to finance their activities (https://www.cnbc.com/2019/07/15/treasury-secretary-mnuchin-will-hold-a-news-conference-on-cryptocurrencies-at-2-pm-et.html). Many in the U.S. congress do not trust FB, thanks in no small measure to FB's past record of rush to profit from selling user data - with little concern about user privacy - to any third party and without any regard to how those third parties may use such data.(https://www.vanityfair.com/news/2019/07/house-facebook-cryptocurrency-libra-moratorium). Already there is lot of political chatter in the US congress about breaking up FB by modifying the anti-trust legislation. The near bi-partisan mistrust in FB was very much on display during recent appearances before congressional oversight committees by head honchos from FB and Libra (https://www.c-span.org/video/?465293-1/facebook-ceo-testimony-house-financial-services-committee and https://www.c-span.org/video/?462708-4/facebook-digital-currency). Given the bi-partisan political opposition and many regulatory hurdles, it appears that a Libra cryptocurrency combined with Callibra's digital finance services as FB has envisioned[9] may have difficulty to go global. However, FB's efforts to create a stablecoin will pave the way for others to create a global stablecoin.

Policy Implications

We live in a world where nearly each country has its own sovereign currency, i.e., its native currency is either created by or under control of a central authority (central bank[10]) representing that country's government. Over 90% or more of this fiat currency is in the form of digital money. Compare that with cryptocurrency, the other kind of digital money; its share is but a tiny fraction of all the digital money in the world[11]. And yet, all over the globe, many central banks and their governments appear to be concerned about emergence of cryptocurrencies. So, what is it about cryptos that worries governments?

For the first time in the long history of money, someone[12] has been able to create a private digital currency with a "decentralized peer-to-peer payment network that is powered by its users with no central authority or middlemen" (https://bitcoin.org/en/faq#what-is-bitcoin). Central authority here refers to central banks who in turn represent nation states.

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[9] According to the Libra white paper (https://libra.org/en-US/wp-content/uploads/sites/23/2019/06/LibraWhitePaper_en_US.pdf) its aim is to become the "internet of money," a reliable digital global currency with almost zero volatility. They really are describing a global stablecoin. However, given FB's checkered past of monetizing every aspect of user usage and behavior, one must examine how well Libra describes the that have been described in Libra's white paper for it to earn the truly global stabelcoin

[10] In the U.S. the Federal Reserve – the U.S. Central Bank – has the sole authority to issue new and replacement currency notes and coin. According to Federal Reserve, depending on the demand for new cash, the ratio of new to replacement is roughly 1 to 7. Replacement currency is just reissuing worn out currency notes and coins that is no longer in circulation. (https://www.federalreserve.gov/faqs/how-does-the-federal-reserve-board-determine-how-much-currency-to-order-each-year.htm).

[11] Total digital currencies of the world amount to over $70+ trillion dollars. Based on CIA's "The World Fact Book" (https://www.cia.gov/library/publications/the-world-factbook/rankorder/2215rank.html), the total broad money is $80 trillion. Assuming 90% of that is in digital form (https://www.marketwatch.com/story/this-is-how-much-money-exists-in-the-entire-world-in-one-chart-2015-12-18), one gets a figure of $72 trillion. Compare that with around $185 billion capitalization of Bitcoin and another roughly $65 billion for all other cryptocurrencies of the world (https://www.banks.com/articles/cryptocurrency/cryptocurrency-market-capitalization/). In brief, cryptocurrencies are a very small fraction (0.35 %) of all the digital money in the world.

[12] Satoshi Nakamato, Bitcoin White paper (http://bitcoin.org), 2009

With no capital control, cryptocurrency networks can operate without any borders; transferring something of value for very small fees between any two parties who may not have to trust each other and without delays associated with bank-to-bank transfers.

Rise of cryptocurrencies highlight what is lacking in fiat currencies: need for peer-to-peer payment services that are cheap, fast, secure and guaranteed.

Additionally, the potential for adoption of cryptocurrencies by people across the globe remains high. Especially when cryptocurrencies are combined with social network platforms that span globally. Such social network platforms can be in the form of simple to use messenger apps on mobile phones. Thus, sending or receiving money could become as easy as sending emails or text messages over mobile phones; essentially replacing outdated and costly payment services that are based on existing financial infrastructure of centralized control.

The emergence of stable coins such as FB's announcement of its cryptocurrency Libra highlights many of the main concerns that Central Banks have with cryptocurrencies. FB's global spanning social network platforms (over 2.4 billion active users across the globe) when combined with its stablecoin Libra could lead to rapid adoption of a *private cryptocurrenc*y by people all over the world. Fast and efficient cross-border payments and remittances with very small fees will benefit both the senders and recipients.

There are over 1.4 billion unbanked individuals across the globe (need citation). Many of them have access to mobile phones (Figure 8), where SIM cards in these phones can vouch for user's identity. A stablecoin technology combined with a social network app on mobile phones could help these individuals with fresh opportunities to actively participate in local, regional and global commerce, allowing them to be part of the UN[13] and G-7 goal of helping create inclusive

[13] https://www.undp.org/content/undp/en/home/2030-agenda-for-sustainable-development/prosperity/development-planning-and-inclusive-sustainable-growth.html

economies (http://www.oecd.org/tax/tax-policy/a-fiscal-approach-for-inclusive-growth-in-g7-countries.pdf).

Are there any downsides to cryptocurrencies in general and stablecoins combined with social network platforms in particular? There are a few mentioned below. At the same time, it is important to note that a *public stablecoin* issued by a central bank could help mitigate many of these potential problems in the future.

Without use of cash, payment services will be exclusively offered by private entities, which could lead to private network monopolies. Public stablecoins can mitigate private monopolies.

Private digital currencies could result in helping tax evasion and criminal activity. Public stablecoin could neutralize these undesirable and avoidable effects.

When unregulated free social network apps replace archaic payment services currently available, use of such apps will generate massive amounts of personal and financial transactions data that could be monetized without user permission. Free social apps and associated very small transaction fees make the users the ultimate commodity that could be used by unregulated private entities for whatever purpose they see fit in their own interest with no regard for user privacy.

Many of these stablecoins are backed by fiat currencies (for example the U.S. Dollar, Japanese Yen, Euro etc.). If public loses its confidence in a private institution issuing stablecoins, there could a run on these stablecoins - trying to convert them into fiat currencies – which in turn could lead to run on the regulated financial institutions creating potentially a global financial crisis.

One of the ways the central banks could mitigate some of these problems is to figure out how to regulate private online social network apps that also offer payment services. Another and potentially more stable option are for Central banks to issue their own digital coins and create new payment services. Additionally, with higher degree of financial inclusion, central banks will be able to track in near real-time distribution of money and growth of debt at greater degree of granularity[14]. However, such digital money tracking might conflict with the degree of anonymity[15] that is guaranteed with cash money. Central banks will have to address these issues of managing a very delicate balance between guaranteeing anonymity and intrusive surveillance.

In brief, private or public stablecoins offer great opportunities of creating inclusive economies and yet pose challenges to citizen privacy and potential for creation of a surveillance state. Resolving some of the negative aspects could make stablecoin technologies a net benefit for people all over the world.

Conclusion

The current payment services with high fees and uncertain delivery times are a drag on cross-border payments. Deep penetration of mobile phones in many remote regions of the world may have opened up new avenues of reaching the unbanked population (Map 6). If inclusive economy is the goal for the low- and middle-income countries, then these countries may want to give figure out a way to allow international flow of digital currencies across their borders. Even some of the high-income countries have large percent of their population who are unbanked but do have access to mobile phones. In that case, creating digital versions of their fiat currencies that could be transferred over mobile phones within their borders would help their unbanked population.

[14] https://www.imf.org/~/media/Files/Countries/ResRep/EUO/cryptocurrencies-and-monetary-policy-kingscollege-2019.ashx

[15] China's central bank, People's Bank of China (PBOC) has expressed its desire to issue its own digital money – the crypto renminbi. One of the potential downsides of this endeavor might be use of digital money as a surveillance tool to spy on people. One of the features of such digital currency is something called "Controllable Anonymity." (https://www.technologyreview.com/f/614711/china-says-its-digital-currency-will-have-controllable-anonymitybut-who-will-control-it/) One can only imagine its potential as a surveillance tool on users of renminbi.

Given the importance of remittances to LMI countries, G20 meeting back in 2014 issued a report in which one of the goals was to facilitate reduction of transfer fees (G20 Plan to Facilitate Remittance Flows, 2014). May be a creation of a global digital currency/stablecoin backed by major fiat currencies could help. This type of global stablecoin would not only help remittances and cross-border payments, reduction of transaction fees would be a huge boost to international commerce and pave the way towards a more equitable and inclusive global economy.

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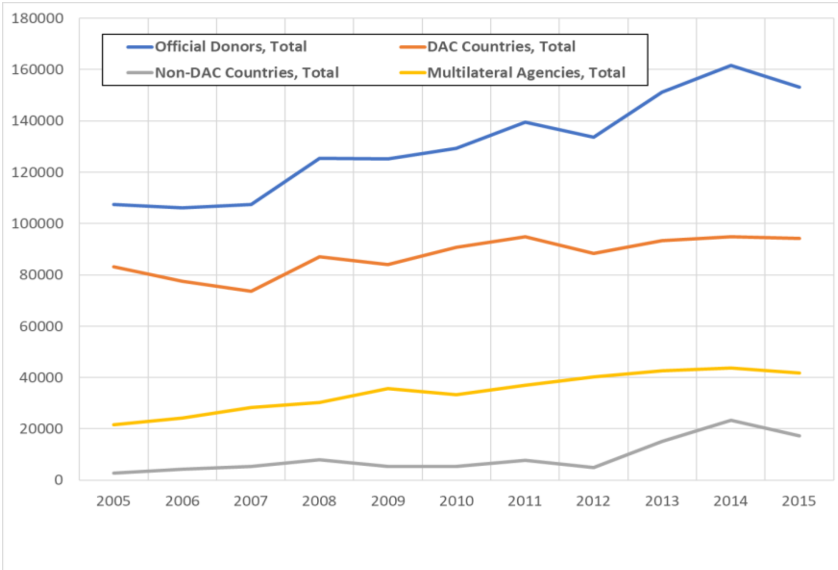


Chart 1: Development Aid

DAC = Development Assistance Committee. The following 30 countries constitute DAC

Australia, Austria, Belgium, Canada, the Czech Republic, Denmark, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Japan, Korea, Luxembourg, The Netherlands, New Zealand, Norway, Poland, Portugal, Slovak Republic, Slovenia, Spain, Sweden, Switzerland, the United Kingdom, the United States and the European Union

Source: OECD Query Wizard (https://stats.oecd.org/qwids)

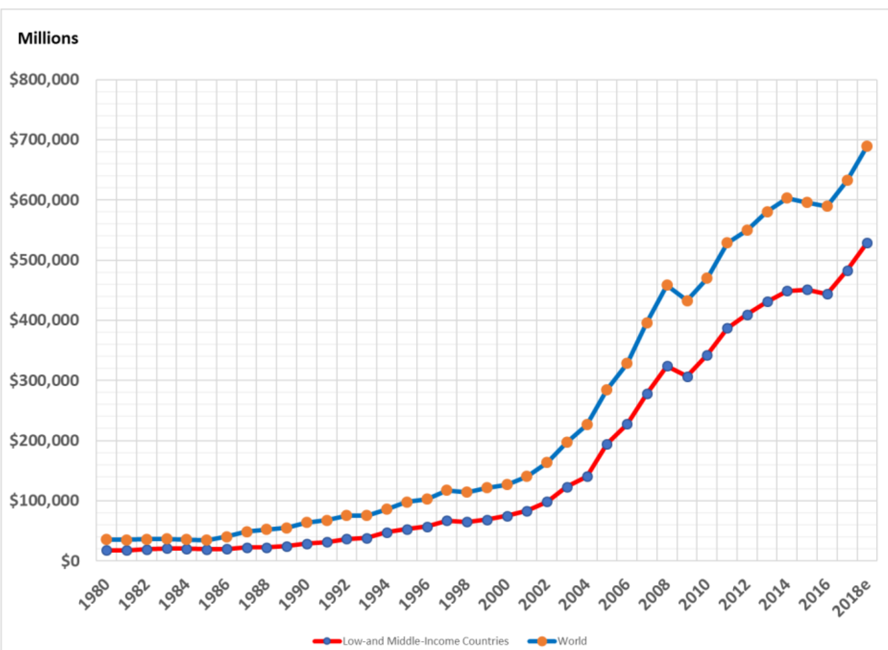
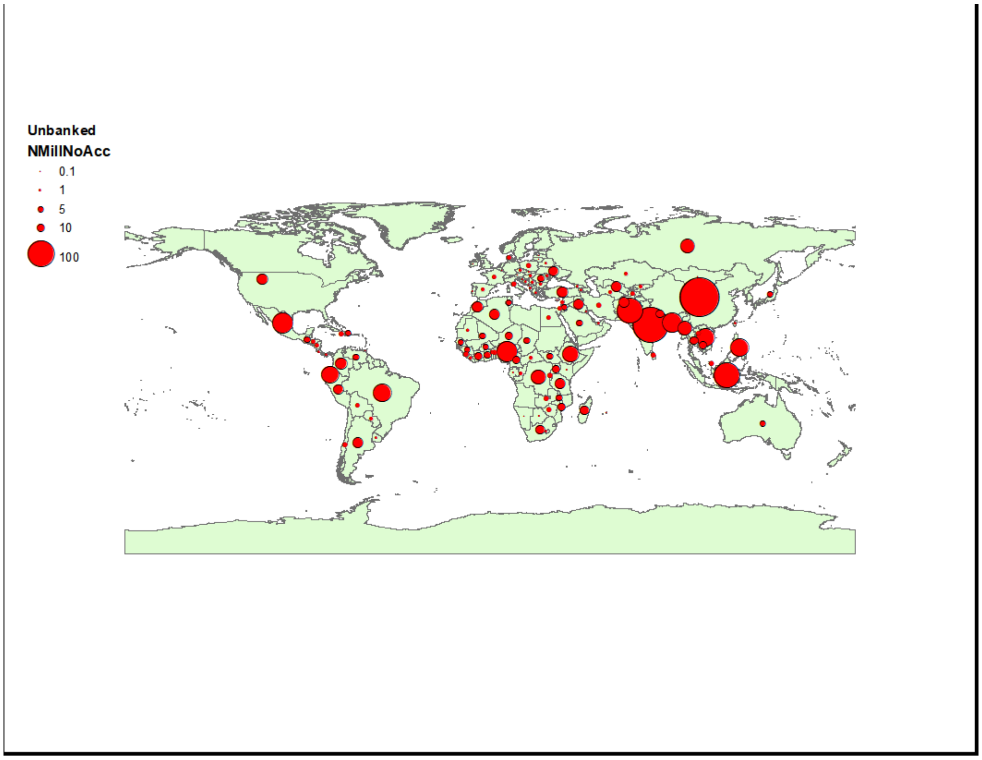


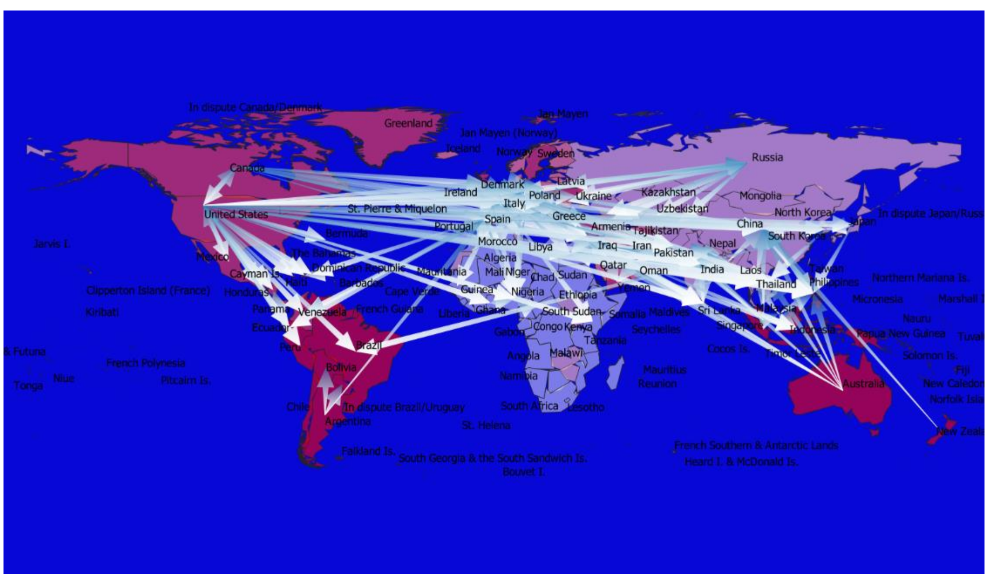
Chart 2: Remittance flows Global and to LMIC

Source: OECD Query Wizard (https://stats.oecd.org/qwids)



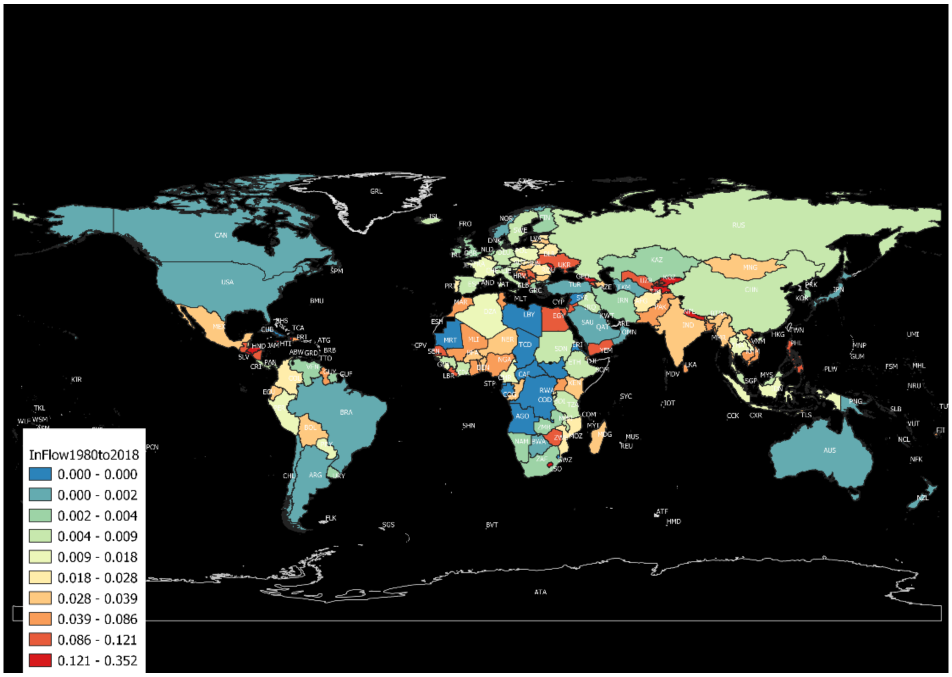
Map 1. Unbanked Population (in millions) by countries 2017

Data Source: The Global Findex Database 2018 (www.globalfindex.worldbank.org)



Map 2: Remittance Flows

Data Source: The World Bank, Remittance Prices Worldwide, (http://remittanceprices.worldbank.org)



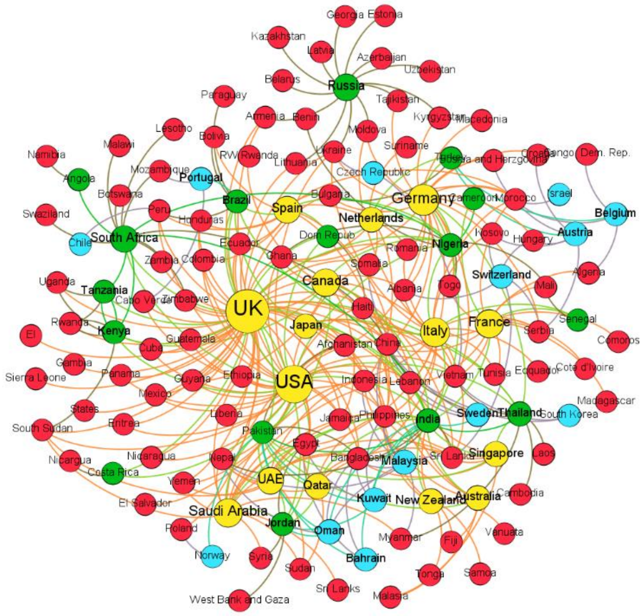
Map 3: Share of Remittances as nation's GDP in 2018

Data Source: The World Bank, Remittance Prices Worldwide (http://remittanceprices.worldbank.org)

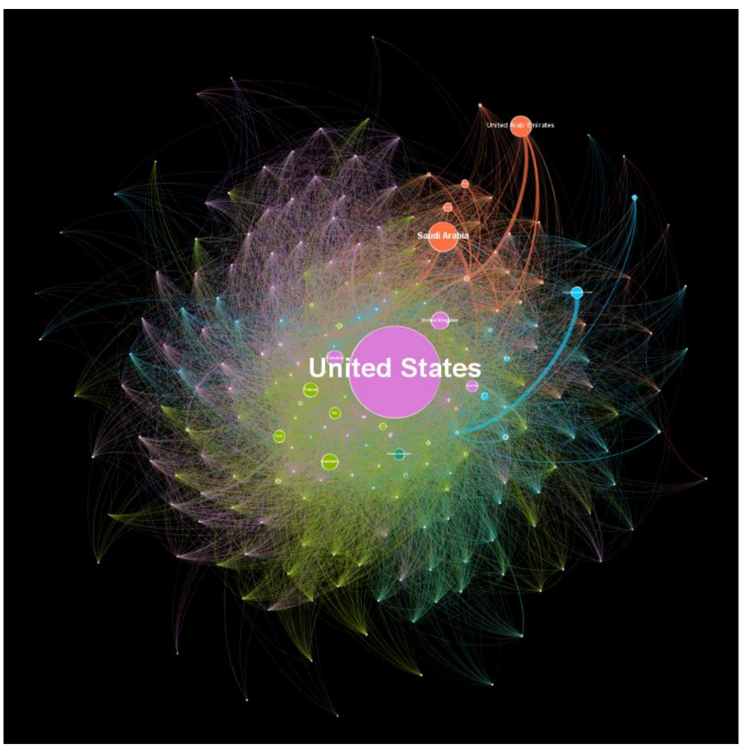
Country	Remittances as % GDP 2018
Tonga	35.2%
Kyrgyzstan	33.6%
Tajikistan	31.0%
Haiti	30.7%
Nepal	28.0%
El Salvador	21.1%
Honduras	19.9%
Comoros	19.1%
Palestine Territories: Gaza Strip	17.7%
Samoa	16.1%
Moldova	16.1%
Jamaica	15.9%
Kosovo	15.8%
The Gambia	15.3%
Lesotho	14.7%
Marshall Is.	13.0%
Lebanon	12.7%
Cape Verde	12.3%
Georgia	12.2%
Guatemala	12.1%
Armenia	12.1%
Liberia	12.0%
Yemen	11.7%
Egypt	11.6%
Ukraine	11.4%
Montenegro	11.4%
Nicaragua	11.2%
Bosnia & Herzegovina	11.0%
Jordan	10.4%

Table 1: Countries with remittances contributing 10% or more of their GPD in 2018

Data Source: The World Bank, Remittance Prices Worldwide, (http://remittanceprices.worldbank.org)

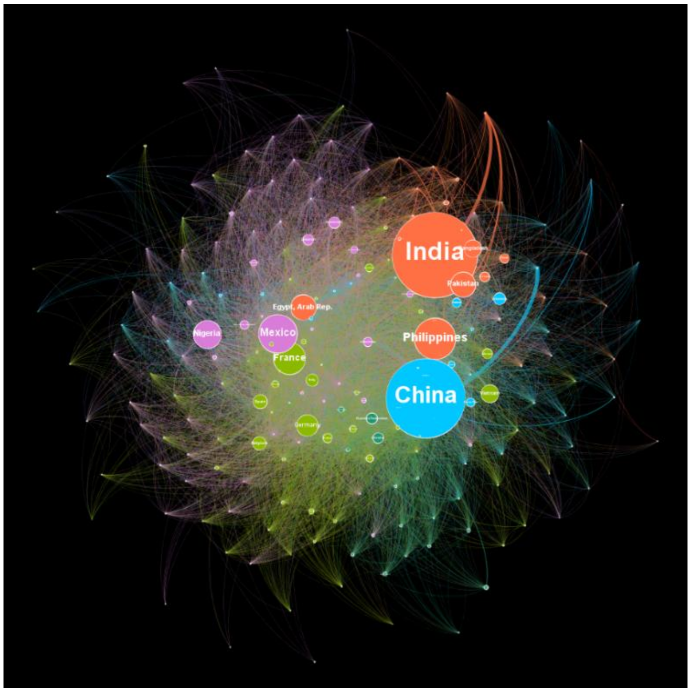


Net-Fig 1: Network topology of Remittance "Donor" and "Recipient" countries.



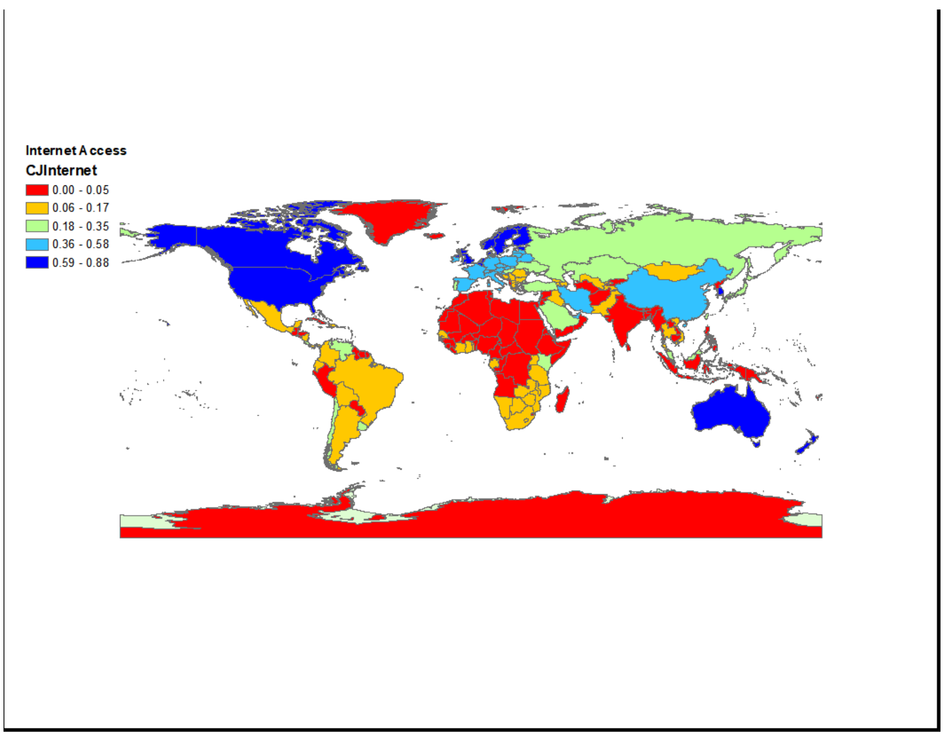
Network Diagram 1: Remittance "Donor" countries

Data Source: The World Bank, Remittance Prices Worldwide, (http://remittanceprices.worldbank.org)



Network Diagram 2: Remittance “Receiver” Countries

Data Source: The World Bank, Remittance Prices Worldwide, (http://remittanceprices.worldbank.org)



Map 5: Share of population with internet access (Data source: World Bank, 2015)

Fig 6: Libra Association Members: Of the original 28 members, 7 have left the association including some bit players such as PayPal, ebay, Visa and Master Card

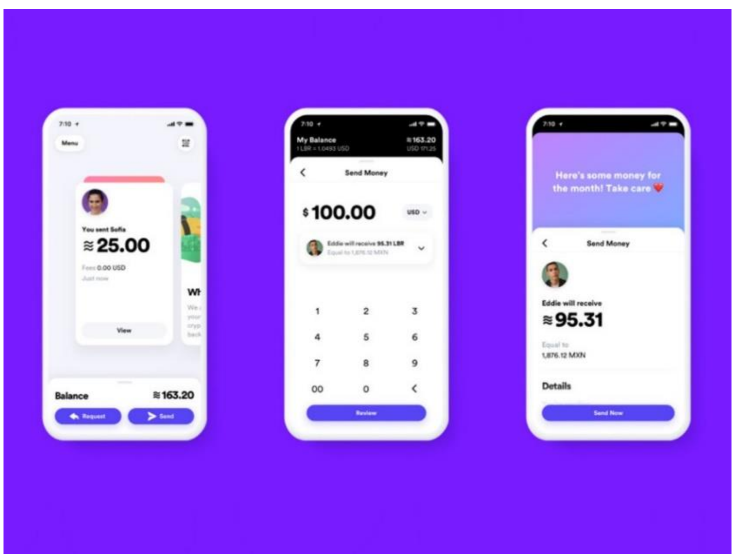


Fig 7: FB's fully owned subsidiary Calibra will implement its version of digital wallets as a stand-alone app that can be used for financial and payment services. FB likes to point out that Calibra wallet can be integrated into FB social network apps for ease of use, faster delivery times, its safe and secure.

Map 6: Unbanked who have access to mobile phones

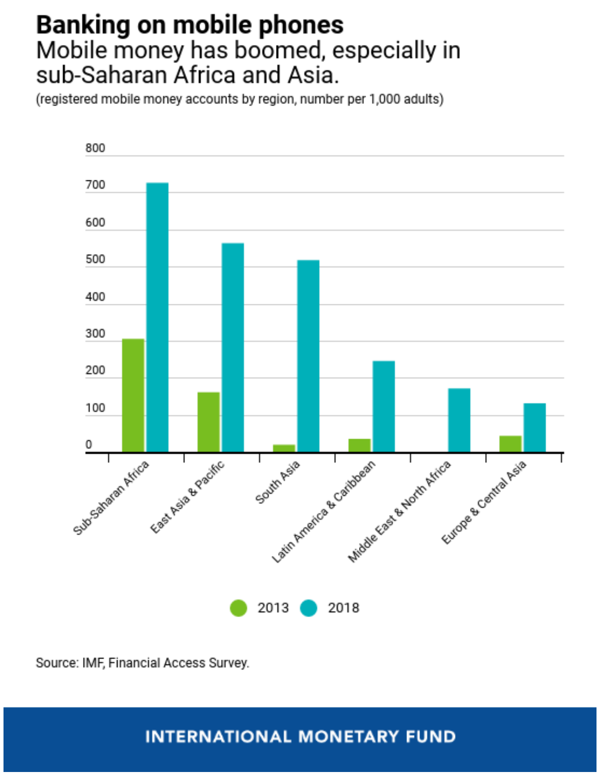


Figure 8: Potential for mobile banking in select countries